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Conservation tillage and nutrient management effects on productivity and soil carbon sequestration under double cropping of rice in north eastern region of India

Gulab Singh Yadav[a,b,*], Rattan Lal[b], Ram Swaroop Meena[b,c], Subhash Babu[d], Anup Das[d], S.N. Bhowmik[a], Mrinmoy Datta[a,e], Jayanta Layak[d], Poulami Saha[a]

[a] ICAR Research Complex for NEH Region, Tripura Centre, Tripura, India
[b] Carbon Management and Sequestration Centre, The Ohio State University, Columbus, USA
[c] Institute of Agricultural Sciences, Banaras Hindu University,Kabir Colony, Varanasi, Uttar Pradesh, 221005, India
[d] ICAR Research Complex for NEH Region, Umiam, Meghalaya, India
[e] College of Agriculture, Tripura University, Tripura, India

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ABSTRACT

The rice (*Oryza sativa*)–rice system (RRS) is the most important agricultural production system, and it provides staple food, income, employment, and livelihoods to millions of farmers in the Indian sub- continent, especially in the eastern and north eastern region (NER) of India. However, soil degradation, due to loss of soil carbon (C) and nitrogen (N) pools, is declining the productivity of RRS and threatening the region's food security. Intensive tillage along with improper residues and nutrient management practices are among the reasons of the loss of soil C and N pools and decline in rice productivity. Therefore, a 3-year (2013-15) field study was conducted to evaluate the effects of tillage, residues and nutrient management practices on productivity, soil C and N sequestration in RRS at the Indian Council of Agricultural Research (ICAR)-Research Complex for the North Eastern Hill (NEH) Region, Lembucherra (52 m, above sea level), Tripura, India. The experiment consisted of five combinations of tillage [conventional tillage (CT), reduced tillage (RT) and no-till (NT)], residue [30% rice residue incorporation (RI) and/or residue retention (RR)] and nutrient management practices [inorganic, organic (FYM-farmyard manure, GLM-green leaf manuring) and biofertilizers] in wet (WR) and dry season rice (DR). Results revealed that RT along with improved plant nutrient management (IPNM) comprising 25% N (20 kg N) through GLM + 60 kg N, 9 kg phosphorus (P), 17 kg potassium (K), 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} through fertilizer + cellulose decomposition microorganism and RR in WR produced significantly higher grain yield (5.15 Mg ha^{-1}) as compared to other treatments. However, the DR transplanted under CT + integrated nutrient management (INM) comprising 25% N through FYM and 75% N and remaining P and K (after deducting quantity supplied by FYM) through inorganic fertilizer + RI produced more grain (5.1–5.3 Mg ha^{-1}), straw (7.0–7.2 Mg ha^{-1}), root (1.3–1.4 Mg ha^{-1}) and total biomass (13.4–13.9 Mg ha^{-1}) yield than that of the farmers' practice (FP) and other treatment combinations, across the years. The highest system productivity of RRS was recorded under T_3 (RT + IPNM + RR in WR and CT + INM + RI in DR). Therefore, the highest biomass, C, and N were also recycled in the system through the same treatment combinations. Soil under T_3 had a lower bulk density (ρ_b), the highest soil organic carbon (SOC)/N concentration, pool, accumulation, sequestration, C retention efficiency, soil microbial biomass C and dehydrogenase activities than other treatments. A total amount of 1.30 Mg C ha^{-1} was accumulated under soils of T_3 with the rate of SOC sequestration of 427.9 kg ha^{-1} yr^{-1} under RRS. Thus, adopting RRS under RT/NT with INM/IPNM and effective residue recycling is recommended for enhancing the system productivity, C and N sequestration in paddy soils of the NER of India.

* Corresponding author at: ICAR Research Complex for NEH region, Lembucherra, Tripura-793103, India.
E-mail addresses: gulabicar@gmail.com, yadav.77@osu.edu, gulab.iari@gmail.com (G.S. Yadav).

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1. Introduction

The rice (*Oryza sativa*)–rice system (RRS) is the principal agricultural production system in Asia, occupying about 24 million ha (M ha) land, of which 6.0 and 2.4 M ha are in India and Bangladesh, respectively (Ladha et al., 2003; Bhuiyan et al., 2004; Haque et al., 2016). It is the second most dominant cropping system after the rice–wheat (*Triticum aestivum*) system in India (Mohanty et al., 2013). The RRS is mainly practiced in regions with assured rainfall and irrigation, and where the winter temperature does not fall below the critical threshold of 10 °C. Two rice crops are grown in these areas in a year viz., wet season rice (WR) (July–December) and dry season rice (DR) (January–June). While, the WR is mostly rainfed; DR is entirely dependent on irrigation and residual moisture (Mohanty et al., 2013). The RRS is fundamental to income, employment, and livelihoods of millions of rural and urban poor in the Indian sub-continent (Das et al., 2014). But now, there are reports of yield decline in a number of long-term RRS experiments (Dawe et al., 2000; Mohanty et al., 2013). Therefore, annual growth rates of rice grain production have not kept pace with that of the population (Haque et al., 2016). Thus, the region's food security is threatened by the emerging challenges of declining productivity and soil degradation under RRS due to loss of soil carbon (C) and nitrogen (N) pools (Das et al., 2014; Lal, 2015). In addition, climate change is also affecting the rice production and has adversely impacted the food security (IWMI, 2007). Indeed, for every 1 °C rise in mean temperature, there is a corresponding 7% decline in rice yield (IWMI, 2007). To keep pace with population growth, rice yields in both the irrigated and rainfed lowland environments must increase by 25% over the next 20 years (Ramesh and Chandrasekaran, 2004). In this context, developing the integrated crop management technologies and restoring C and N pools in soils are the viable alternatives for increasing and sustaining crop yields.

Within the terrestrial ecosystems, paddy (rice) soils are important factors affecting the global C cycle (Mandal et al., 2008; Rajkishore et al., 2015). But, most of the rice areas in India, China and other South and South East Asian countries are blamed for source of intense greenhouse gas (GHG) emissions (Ma et al., 2016; Mi et al., 2016; Meena and Meena, 2017). However, paddy soils retain higher amounts of C and N among all terrestrial ecosystems (Liu et al., 2006; Xie et al., 2007). Further, soil organic matter (SOM) is preferentially accumulated in continuous rice systems as a result of submerged and anaerobic conditions (Majumdar et al., 2008; Bhattacharyya et al., 2013). Thus, slower decomposition of SOM and higher net productivity of submerged paddy soils lead to net C and N accumulations (Majumdar et al., 2008; Bhattacharyya et al., 2013). Small incremental changes in the soil organic carbon (SOC) pool could result in significant impacts on the atmospheric C concentration (Stockmann et al., 2013; Mi et al., 2016). In contrast, intensive tillage, residue removal, and improper nutrient management in rice ecosystems can deplete SOC and N pools and disturb their dynamics (Majumdar et al., 2008; Bhattacharyya et al., 2013). Therefore, C and N sequestration by using improved soil and crop management practices is required for enhancing the system productivity and soil quality of rice- based ecosystems. A sound knowledge of soil C and N cycling in the RRS is needed for a better understanding of these processes (Sainju et al., 2008).

Agricultural practices for enhancing SOC must either increase organic matter (OM) inputs to the soil (Yadvinder-Singh et al., 2005; Jat et al., 2009), decrease decomposition of SOM and oxidation of SOC, or a combination of both (Paustian et al., 2000; Prasad et al., 2016). Reducing tillage intensity and retaining residues can enhance the C and N pools in soils of diverse cropping systems (Lal and Kimble, 1997; Bhattacharyya et al., 2013; Sharma et al., 2014). In general, SOC increases with use of organic amendments, and conservation tillage (e.g., reduce tillage-RT and no-till-NT), and crop rotation (Bhattacharyya et al., 2007; Purakayastha et al., 2008; Tong et al., 2009). Conservation agriculture (CA) consisting of RT/NT, retention of crop residues can accumulate soil C and N in the temperate, tropical and Mediterranean environments (West and Post, 2002; Gonzalez Sanchez et al., 2012; Prasad et al., 2016; Meena et al., 2017).

CA is practiced in various forms over 157 M ha worldwide (Kassam, 2014; Haque et al., 2016), but mostly in large and mechanized farms in rainfed and limited irrigated areas (Haque et al., 2016). Experimental evidences from rice–wheat and maize (*Zea mays*)–wheat systems suggest that CA-based alternative tillage and crop establishment methods can produce both immediate e.g., reduced production cost, stabilized crop yields, and improved water productivity (Hobbs, 2007; Ladha et al., 2009; Jat et al., 2009; 2013), and long-term benefits like improved soil quality (Gathala et al., 2011; Kienzler et al., 2012). However, there is much less adoption of CA in RRS, practiced predominantly by smallholder farms (Johansen et al., 2012). The common method of rice establishment in Asia involves mostly hand transplanting following repeated/intensive tillage and soil puddling (Haque et al., 2016) with some amount of rice residue incorporation and minimum amount of fertilization Das et al., 2014). This practice can exacerbate the loss of soil C and N Das et al., 2014; Singh et al., 2016); destroy soil aggregates (Gathala et al., 2011; Kienzler et al., 2012); disrupt capillary pores (Hobbs, 2007); disperse clay particles that can form an impermeable clayey layer on the surface of coarse-textured soil (Johansen et al., 2012); and create a plow pan (compacted layer) that impedes root penetration and causes water flooding in areas of heavy rainfall (Johansen et al., 2012; Haque et al., 2016). Based on a study in Bangladesh, Haque et al. (2016) reported significant advantages in terms of increased yields and higher income from the minimum tillage over the conventional tillage in RRS.

The resource rich North Eastern region (NER) of India located in eastern Himalayas has a population of ~47 million on a geographical area of 26.2 M ha (Das et al., 2015). The NER is characterized by diversified agro- climatic conditions and topography (hills and valley), and about 70% of the region is hilly (Yadav et al., 2013a; Meena and Yadav, 2014). The plain areas and valleys surrounded by hills are used for rice cultivation (Das et al., 2014). Rice is a staple food of the NER, and is cultivated on about 3.5 Mha, which account for almost 8% of the land area and 6.5% of the country's rice production (Choudhary et al., 2016). The RRS is the most pre-dominant system in the region (Yadav et al., 2013a). The major concern is that, despite the favorable edaphic and climatic conditions; the rice yield in the region hardly exceeds 2 Mg ha^{-1} as against national average of 2.8 Mg ha^{-1} in India and 4.42 Mg ha^{-1} in neighboring Bangladesh (Das et al., 2015). Food security in the NER entirely depends on rice because no other cereals are grown by the farmers due to food habits and agro-climatic situations (Das et al., 2014; Yadav et al., 2017). Thus, self- sufficiency in food grains in the region can be achieved only through enhancement in productivity of RRS (Das et al., 2015). This is possible by developing improved resource- effective technologies, and management and restoring soil's C and N pools. The CA technologies with some modification [e.g., RT, NT, INM (integrated nutrient management), use of micro nutrient, and residue retention] in rice–wheat and maize–wheat cropping system have been proven to increase productivity and the soil health under irrigated conditions in the Indo-Gangetic Plains (Singh et al., 2016). However, the research information on the impact of CA on productivity and sustainability of RRS under the rain-fed NER ecosystems is scanty (Nagarajan et al., 2013). Therefore, the present study was conducted to assess the effects of different tillage, nutrient and residue management options on system productivity, soil C and N pools of RRS under the climatic conditions of the NER of India. The experiment tested the hypothesis that conservation tillage with judicious management of nutrient and crop residues enhances the system productivity, and increases C and N sequestration in soils under RRS.

2. Materials and methods

2.1. Experimental site

A 3-year study (2013–2015) on RRS was conducted in the lowland rice field at the Agronomy experimental farm (*Cocotilla* farm) of the Indian Council of Agricultural Research (ICAR)-Research Complex for North Eastern Hill (NEH) Region, Tripura Centre, Lembucherra, Tripura (W), India (23^{0}54′24.02”N and 91^{0}18′58.35”E, and altitude of 52 m above mean sea level). Prior to initiation of the actual experiment, one wet season rice (WR) crop was grown (2012) uniformly with traditional farmers practice (FP) for homogenization of soil fertility in the experimental field. The soil (*Typic Kandihumults*) of the experimental field is clay loam, deep and free from gravels and hardpan. Base line soil samples from 0 to 20 cm depth were collected and analyzed before initiation of the experiment. The soil had 10.2 g kg^{-1} of SOC concentration as determined by the dry combustion method (Nelson and Sommers, 2005) using a TOC analyzer (ElementarVario Select, Germany), 1.01 g kg^{-1} total soil nitrogen (TSN) by the Kjeldahl digestion method (Bremner, 1960), 9.5 mg kg^{-1} available P and 295.7 mg kg^{-1} available potassium (K) by Prasad et al. (2006). The soil pH was 5.1 (1:2.5, soil and water ratio) (Glendon and Dani, 2002) and bulk density (ρ_b) 1.33 Mg m^{-3} by the core method (Blake and Hartge, 1986).

2.2. Weather information

The long-term average annual rainfall over 30 years of the experimental site is 2200 mm, with > 65% received between June and September. However, cumulative rainfall during the period of investigation in the year 2013, 2014 and 2015 were 2056.8, 1821.1 and 2397.3 mm, respectively. The average monthly distribution of rainfall, temperature and relative humidity (RH) of three year (2013-15) are given in Fig. 1.

2.2.1. Wet season rice

A calendar year was divided into two parts: from January to June (dry season) and July to December (rainy or wet season). The WR and DR were grown during the rainy and dry seasons, respectively. The WR crop was grown in nursery during June and transplanted in the main field during July. The crop took about four month from transplanting to harvesting. The rainfall received during this season had a large variation over the three year period. The season received the maximum (1298.4 mm) rainfall during 2015 and the minimum (758.5) in 2013. There was also variation in monthly distribution of rainfall. In 2013, September month received the maximum amount (266.8 mm), while October received the minimum (115.1 mm). However, the trend was different in 2014 and 2015, when August (392.6 mm) and July (610.0 mm) received the maximum rainfall and October received the lowest (96.9 and 124.8 mm) in both years. Rainfall received in June and July is more important for the raising of nursery and transplanting

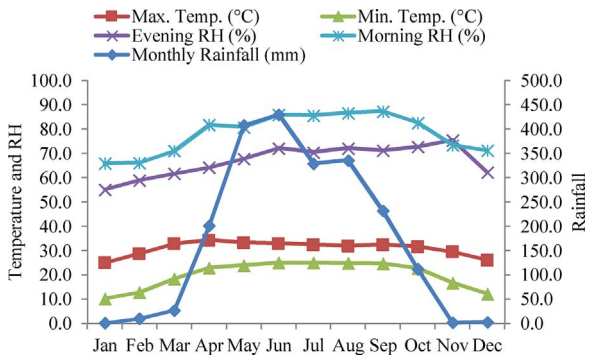


Fig. 1. Average meteorological observation of experimental site from 2013 to 2015.

of rice. The average maximum and minimum temperature did not vary across the years, and ranged from 30.6 to 30.9 °C and 20.9–21.6 °C, respectively. But both the maximum and minimum temperatures decreased from July to December during all the years. The monthly variations of some weather parameters are presented in Table 1.

2.2.2. Dry season rice

The dry season rice was transplanted during winter (January) and harvested in summer (May–June). The amount of rainfall received and temperature variation from January to June was measured for the entire season. This is an irrigated crop, but a significant amount of rain was received from April to June in every year, which reduced the requirement for number of irrigations. There was a large variation in total rainfall received during this season, and across the years. During dry season, the maximum rainfall (1298.3 mm) was received in 2013 and the minimum (836.1 mm) in 2014. The highest monthly rainfall ranged between 99.4–369.0 mm in April, 236.3–714.9 mm in May and 400.0–472.6 mm in June in different years (Table 1). There was a little rainfall at the beginning of the rice season during January (0–3.4 mm) and February (6.2–15.2 mm). The year 2014 was relatively dry compared to years 2013 and 2015. The maximum temperatures during June to September ranged between 25.1 and 33.0 °C and the minimum temperatures between 10.3 and 25.0 °C.

2.3. Experimental design and crop management

The experiment was laid out in a randomized complete block design (RCBD) with three replications. The experiment consisted of five combinations of tillage, residue and nutrient management practices as follows:

(1) **T_1**: Conventional tillage (CT) + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation (RI) + farmyard manure (FYM) 5 Mg ha^{-1} once in two years in WR and CT + 40 kg N and 9 kg P ha^{-1} + 30% RI in DR;
(2) **T_2**: CT + 80 kg N, 18 kg P and 33.3 kg potassium (K) ha^{-1} (RDF) + 30% RI in WR and CT + RDF + 30% RI in DR;
(3) **T_3**: Reduce tillage (RT) + integrated nutrient management (INM) comprising 25% N through FYM and 75% N & rest of P & K (after deducting quantity supplied through FYM) through inorganic fertilizer + 30% residue retention (RR) in WR and CT + INM + 30% RI in DR;
(4) **T_4**: RT + improved plant nutrient management (IPNM) comprising 25% N through green leaf manure (GLM) + 60 kg N, 9 kg P, 17 kg K, 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} + cellulose decomposing micro-organism (CDM) + 30% RR in WR and RT + RDF + CDM + 30% RR in DR; and
(5) **T_5**: No-till (NT) + IPNM + CDM + 30% RR in WR and NT + RDF + CDM + 30% RR in DR. The gross and net plot size were 6.0 × 5.0 m^2 and 5.0 × 4.0 m^2, respectively. Detailed description of experimental treatment is also given in Table 2 and about tillage and other cultural operations in Table (Table 3).

2.3.1. Seeding and seeding rate

12 kg of seed was sown for seedling production on an area of 120 m^2 for each crop. Twenty- one–day old rice seedlings of popular high yielding rice varieties Naveen (DR) and Gomatidhan (WR) were manually transplanted at 20 × 20 cm spacing with two seedling hill^{-1}. The DR (winter) season rice nursery was raised during the 1st week of January and transplanted during the last week of January every year. During the WR, the nursery was raised during the 3rd week of June and transplanted during the 1st week of July every year.

2.3.2. Manure and fertilizer management

A full dose of P and K and one split dose of N were applied as the basal application before transplanting of rice. The remaining splits of N

Table 1
Mean monthly meteorological observation during 2013–2015.

Parameters	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2013												
Rainfall (mm)	0.0	9.2	38.8	135.4	714.9	400.0	151.2	223.0	266.8	115.1	0.0	2.4
Max. Temp. (°C)	24.2	30.0	33.9	34.6	31.6	34.2	33.2	32.2	32.6	30.9	29.4	26.8
Min. Temp. (°C)	8.5	12.9	19.1	23.3	22.9	25.2	25.3	24.7	24.7	22.7	16.1	11.8
RH (%)	59.0	60.4	63.5	69.5	70.8	77.6	76.1	77.5	77.0	77.8	72.4	63.3
2014												
Rainfall (mm)	0.0	6.2	21.6	99.4	236.3	472.6	225.9	392.6	269.6	96.9	0.0	0.0
Max. Temp. (°C)	25.2	27.9	32.3	35.9	34.4	32.4	32.7	31.6	31.6	32.1	29.8	25.7
Min. Temp. (°C)	10.1	12.3	17.9	25.1	25.2	25.4	25.4	25.2	24.9	23.3	17.4	13.1
RH (%)	58.0	61.1	67.2	71.3	73.8	79.3	76.5	80.8	79.2	81.0	77.4	69.9
2015												
Rainfall (mm)	3.4	15.2	21.4	369.0	272.5	417.3	610.0	391.8	160.0	124.8	5.6	6.2
Max. Temp. (°C)	25.8	28.4	32.4	32.3	33.8	32.3	31.5	32.2	33.3	31.8	29.5	25.8
Min. Temp. (°C)	12.2	13.4	18.5	20.6	23.9	24.5	24.5	24.8	24.5	22.5	17.0	11.9
RH (%)	64.7	66.5	68.7	78.2	78.6	80.4	81.6	80.2	81.8	74.6	73.8	67.2

were applied as top dressing at 30 days after transplanting, as per the treatments. FYM and GLM were applied at the time of the second plowing in CT and RT, and incorporated into the soil. Under NT, both doses were applied on the surface. Thirty percent rice straw was retained/incorporated in the field as per treatment. The rice biomass (straw + root), FYM and GLM applied in field contained 43.1%, 20.1%, 13.2% of C and 0.56%, 0.50%, 2.30% of N, respectively.

2.3.3. Weed management

Weeds in NT plots were controlled by spraying glyphosate (N-(phosphonomethyl) glycine) @ 5 ml l^{-1} using flat fan nozzle (Knapsac sprayer, model AGM/001). After transplanting, pretilachlor (2-Chloro-*N*-(2,6-diethylphenyl)-*N*-(2-propoxyethyl) acetamide) @ 1200 g (a.i.) ha^{-1} was applied manually two days after transplanting. One manual weeding was also done at 30 days after transplanting of rice during both the seasons to keep the plots weed-free.

2.3.4. Water management

The WR was grown under complete rain- fed conditions, but DR was grown with assured irrigation. However, the need of irrigation of crop was based on the soil moisture content and weather conditions. The DR plots were kept flooded (5 cm of standing water) for the first 2 weeks, followed by irrigation (5 cm depth) at the appearance of cracks on the soil surface till maturity. On average, total irrigation water used in DR ranged between 800 and 900 ha-mm across the treatments. However, the highest amount of irrigation water was required under CT (900 ha-mm), because puddling consumed 200 ha-mm water. In contrast, the lowest amount was required under NT plots (800 ha-mm) due to retention of rice residues on the soil surface. Irrigation water to rice in different treatments was applied at all the critical growth stages so that crops were not affected by moisture stress.

2.3.5. Harvesting, economic yield and biomass measurement

Both WR and DR were harvested at physiological maturity during the 2nd fortnight of November and last week of May to 1st week of June, respectively in all the years. Both the rice crops were harvested manually in such a way that 30% of the rice residue was retained in the field. For yield measurement, a net plot area of 5 × 4 m was harvested and kept on the threshing floor to allow the biomass to dry for 4–5 days. The harvested plants were then weighed, and threshed manually to separate the grains from straw. Subsamples of grains and straw were dried in the oven at 70 ± 1 °C to constant weight. Grain yields were adjusted to 14% moisture content for both WR and DR crops. A 2.1 Mg ha^{-1} of the biomass of previous WR (grown during previous rainy season before initiation of present study for treatment stabilization) was considered as the initial biomass contribution to 1st DR crop (2013).

Root samples were obtained at harvest in both crops at 20 cm depth using a core sampler (5.8 cm height and 5.4 cm diameter). Five hills were randomly selected from each plot. The core sampler was placed on soil keeping the base of the stem in the center of the sampler and then pushed vertically to the desired depth. The core samples with roots and soil were soaked in water for at least 12 h following the procedure described by Bohm (1979). The roots were washed free of soil, and the detritus and the fresh roots were oven-dried at 70 ± 1 °C till constant weight, and the dry biomass was determined and converted into Mg

Table 2
Detailed descriptions of experimental treatments.

SN.	Wet season (*Aman*) rice		Dry season (*Boro*) rice	
	Treatment	Symbol	Treatment	Symbol
T_1	Conventional tillage (CT) + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation (RI) + FYM 5 Mg ha^{-1} once in two years	FP	CT + 40 kg N and 9 kg P ha^{-1} + 30% RI	FP
T_2	CT + 80 kg N, 18 kg P and 33.3 kg potassium (K) ha^{-1} (RDF) + 30% RI	CT + RDF + RI	CT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF) + 30% RI	CT + RDF + RI
T_3	Reduce tillage (RT) + integrated nutrient management (INM) comprising 25% N through FYM and 75% N & rest of P & K(after deducting contributions from FYM) through inorganic fertilizer + 30% residue retention (RR)	RT + INM + RR	CT + INM + 30% RI	CT + INM + RI
T_4	RT + Improved plant nutrient management (IPNM) comprising 25% N through green leaf manure (GLM) + 60 kg N, 9 kg P, 17 kg K, 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} + cellulose decomposing micro-organism (CDM) + 30% RR	RT + IPNM + CDM + RR	RT + RDF + CDM + 30% RR	RT + RDF + CDM + RR
T_5	No-till (NT) + IPNM + CDM + 30% RR	NT + IPNM + CDM + RR	NT + RDF + CDM + 30% RR	NT + RDF + CDM + RR

FP: farmer's practice; FYM; farm yard manure; RDF: recommended dose of fertilizer; INM: integrated nutrient management; IPNM: improved plant nutrient management.

Table 3
Details of the tillage and other cultural operations done in rice–rice system.

Operation	Soil depth (cm)	Objective	Convention tillage (CT)	Reduced tillage (RT)	No-tillage (NT)
Deep spading with the manual labor	15–20 cm	Incorporation of rice residue and weed biomass	1 times	–	–
Shallow spading with the manual labor	10–15 cm	Partial incorporation of rice residue and weed management	3 times	2 times	–
Puddling the manual labor	10–15 cm	Water retention and weed control	2 times	–	–
Glyphosate application	–	Weed control	–	–	Seven days before transplanting
Cellulose decomposing micro-organism (CDM) application	–	Decomposition of surface rice residue and weed biomass	–	Fifteen days before transplanting	Fifteen days before transplanting
Farmers practice (FP)	CT with puddling + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation each crop + farm yard manure (FYM) 5 Mg ha^{-1} once in two years to first crop i.e., wet season rice (WR)				
Recommended of Dose fertilizer (RDF)	80 kg N, 18 kg P and 33.3 kg K ha^{-1} in WR and 100 kg N, 18 kg P and 33.3 kg K ha^{-1} in dry season rice (DR)				
Integrated nutrient management (INM)	25% N through FYM and 75% N & rest of P & k through inorganic fertilizer				
Improved plant nutrient management (IPNM)	25% N through *Gliricidia* spp. Leaves and twigs as green leaf manure (GLM) + 60 kg N, 9 kg P, 17 kg K, 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} + CDM				
Cellulose decomposing micro-organism (CDM)	*Trichoderma viride* @ 4 g l^{-1} water (600 l water ha^{-1}) was applied 15 days before transplanting of rice on rice stubbles				

ha^{-1}.

2.4. Soil sampling and analysis

Soil samples were obtained (500 g composite sample, one sample from each plot) from 0 to 20 cm depth for analyzing the SOC, TSN, soil microbial biomass carbon (SMBC) and dehydrogenase activities (DHA) after completion of three years of the experiment. Total soil C was determined by the dry combustion method (Nelson and Sommers, 2005) using a TOC analyzer (ElementarVario Select, Germany). The SOC was assumed to be equal to the total C with negligible inorganic C concentrations as the soil pH was below 7 (Jagadamma and Lal, 2010). SMBC was estimated by soil fumigation technique (Anderson and Ingram, 1993). Soil samples were analyzed for total N by the Kjeldahl digestion method (Bremner, 1960). Soil DHA was estimated by the procedures described by Tabatabai (1982) by reducing 2,3,5-triphenyl tetrazolium chloride (Casida et al., 1964). Soil bulk density (ρ_b) was determined by the core method (Blake and Hartge, 1986) using cores of 5.8 cm height and 5.4 cm diameter at 0–20 cm depth and oven dried at 105 °C (one sample per plot).

2.5. Computation of C and N pools

Total SOC and N pools (Mg ha^{-1}) in 0–20 cm layers were calculated using the fixed depth (FD) method and Eq. (1) (Lee et al., 2009):

$$M_{C/N} = \rho_b \times D_f \times C_{C/N} \times 10 \quad (1)$$

Where, $M_{C/N}$ is the SOC/N mass per unit area (Mg C/N ha^{-1}), ρ_b is the soil bulk density (Mg m^{-3}), $C_{C/N}$ is the concentration of SOC/N (g kg^{-1}), D_f is the depth of the fixed soil layer (m), and 10 is a product of unit conversion factor (m^2 ha^{-1}, g Mg^{-1}, and kg Mg^{-1}).

Sequestration of SOC/N was computed as per Eq. (2):

$$\text{C sequestered (Mg C/N ha}^{-1}\text{ soil)} = [\text{SOC/N current (Mg ha}^{-1}\text{)} - \text{SOC/N initial (Mg ha}^{-1}\text{)}]/\text{year of experimentations} \quad (2)$$

Carbon retention efficiency (CRE) was calculated by Eq. (3):

$$\text{CRE (\%)} = (\text{SOC final} - \text{SOC initial}) \times 100 \div \text{ECI} \quad (3)$$

SOC final and SOC initial represent SOC (Mg ha^{-1}) in the final and initial soils, respectively, and ECI is estimated cumulative C input (Mg ha^{-1}) to soil between the initial and final year of experimentation.

2.6. Statistical analysis

Data were subjected to ANOVA, to test the significance of the overall differences among treatments by the "F" test. When the 'F' value was found to be significant, the least significant difference (LSD) at $p = 0.05$ was computed to test the significance of the difference between treatment means (Gomez and Gomez, 1984).

3. Results and discussions

3.1. Yield of wet season rice

Grain yield, straw yield, root and total biomass of WR crop were significantly affected by tillage, residue and nutrient management practices (Fig. 2). Cultivation of WR under RT + IPNM + CDM + RR recorded significantly higher grain yield (5.1–5.2 Mg ha^{-1}) as compared to other treatments across the years. Thus, there was a positive impact of RT with improved nutrient management practices on the production of WR crop. The straw, root and total biomass yields of WR also followed the trend similar to that of grain yield. The WR grown under RT + IPNM + CDM + RR recorded about 58% higher grain and straw yields over that of the FP, and about 15.7% higher grain and straw yields over that of CT + RDF + RI. However, grain, straw, root

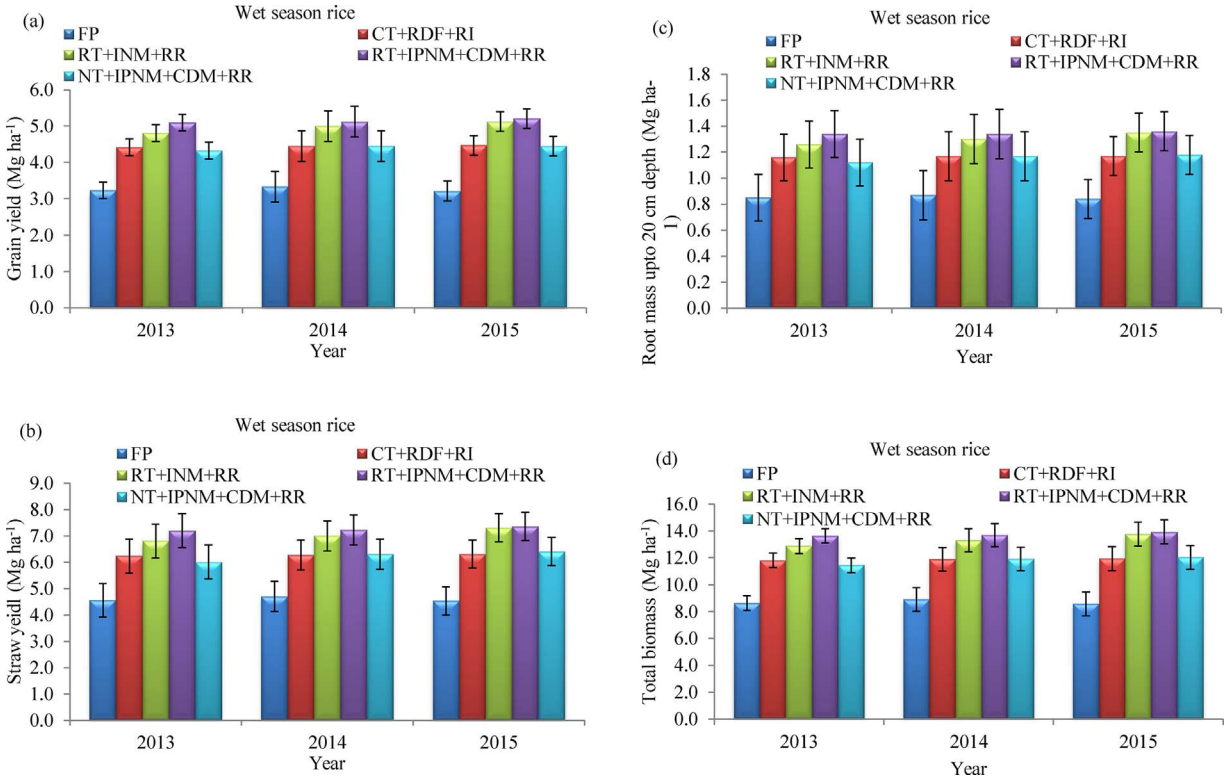


Fig. 2. Effect of tillage and nutrient management on grain (a), straw (b), root (c) and biomass (d) yield of wet season rice under rice–rice system (the vertical error bars indicate LSD at $p = 0.05$%).

FP: Conventional Tillage (CT) + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation (RI) + farm yard manure (FYM) 5 Mg ha^{-1} once in two years (FP); CT + RDF + RI: CT + 80 kg N, 18 kg P and 33.3 kg potassium (K) ha^{-1} (RDF) + 30% RI; RT + INM + RR: Reduce tillage (RT) + integrated nutrient management (INM) comprising 25% N through FYM and 75% N & rest of P & K(after deducting contributions from FYM) through inorganic fertilizer + 30% residue retention (RR); RT + IPNM + CDM + RR: RT + Improved plant nutrient management (IPNM) comprising 25% N through green leaf manure (GLM) + 60 kg N, 9 kg P, 17 kg K, 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} + cellulose decomposing micro-organism (CDM) + 30% RR; NT + IPNM + CDM + RR: No-till (NT) + IPNM + CDM + 30% RR; FP: farmer's practice; RDF: recommended dose of fertilizer.

and total biomass yield differed significantly among CT + RDF + RI and NT + IPNM + CDM + RR. The data for three consecutive years indicated that the conservation tillage with INM/IPMN and 30% RR/RI produced more rice grain yield than that with FP and CT + RDF + RI. These trends indicate opportunity of adopting RT with improved nutrient management in lowland rice cultivation with some yield advantage (Das et al., 2014). Similar grain yield of rice under puddled and minimum puddled treatments was also reported by Singh et al. (2004) and Das et al. (2014). Several researchers (Hammel, 1995; Das et al., 2014) have observed that conversion from CT to CA did not increase crop yields under humid condition, but they did not consider the nutrient management aspects of crops. In clay loam soils, where rainfall and water table are generally high (Das et al., 2014), water percolation is not a problem (similar condition exist in the NER), and puddling has no effect on rice yield (Das et al., 2014). Therefore, a good rice yield with RT + IPNM + CDM + RR in the present study may be attributed to the balanced input of nutrients and residue management practices. Many previous studies on conservation tillage crop establishment have also reported crop yields equivalent to that of the traditional full tillage systems (Ladha et al., 2009; Saharawat et al., 2009). However, most of the previous research evaluated RT/NT vs. CT in wheat and maize (Sharma et al., 2011; Yadav et al., 2015) rather than for the transplanted rice. Sharma et al. (2005) transplanted rice after one wet tillage to puddle the soil (minimum tillage), and reported yield similar to that of transplanted rice after full soil puddling involving several tillage operations. Haque et al. (2016) observed that RT and unpuddled transplanting produced more rice yields across seasons and years over CT system, and reduced the time for land preparation and crop establishment and decreased the cost of production. However, other researchers (e.g., Ladha et al., 2009; Saharawat et al., 2009) also reported that transplanting in unpuddled NT soil increased average rice yields in farmer's fields by 0.3 Mg ha^{-1}. Therefore, the negative effects of RT/NT on crop growth are not observed in wet-seeded flooded rice production systems (Huang et al., 2012). The increase in rice yield in the present study may not be solely due to tillage but because of the combined effect of conservation tillage, and residue and nutrient management practices. Balanced application of nutrients along with micronutrients and residue incorporation has been previously reported to increase the yield of rice (Singh et al., 2009).

3.2. Dry season rice yield

Yields of DR crop (grain, straw and total biomass including root mass up to 20 cm soil depth) were (Fig. 3) significantly affected by tillage, residues and nutrient management practices. But, the response differed as compared to that of WR. Rice transplanted under CT + INM + RI produced more grain (5.1–5.3 Mg ha^{-1}), straw (7.0–7.2 Mg ha^{-1}), root (1.3–1.4 Mg ha^{-1}) and total biomass (13.4–13.9 Mg ha^{-1}) yield than that under FP and other treatment combinations, across the years. The treatment CT + RDF + RI and RT + RDF + CDM + RR did not have significant difference in yield (grain, straw, root and total biomass). However, transplanting of rice under RT + RDF + CDM + RR and NT + RDF + CDM + RR systems caused a yield reduction by 9.3 and 19.1% over that under CT + RDF + RI, respectively. In general, CT with puddling has more impact during the dry season by reducing percolation loss of water and controlling weeds because of low rainfall and deeper water table (Das et al., 2014). The highest yield recorded under CT + INM + RI might be due to the beneficial impact of INM. High plant growth and development and high yield in INM may be attributed to better nutrient availability

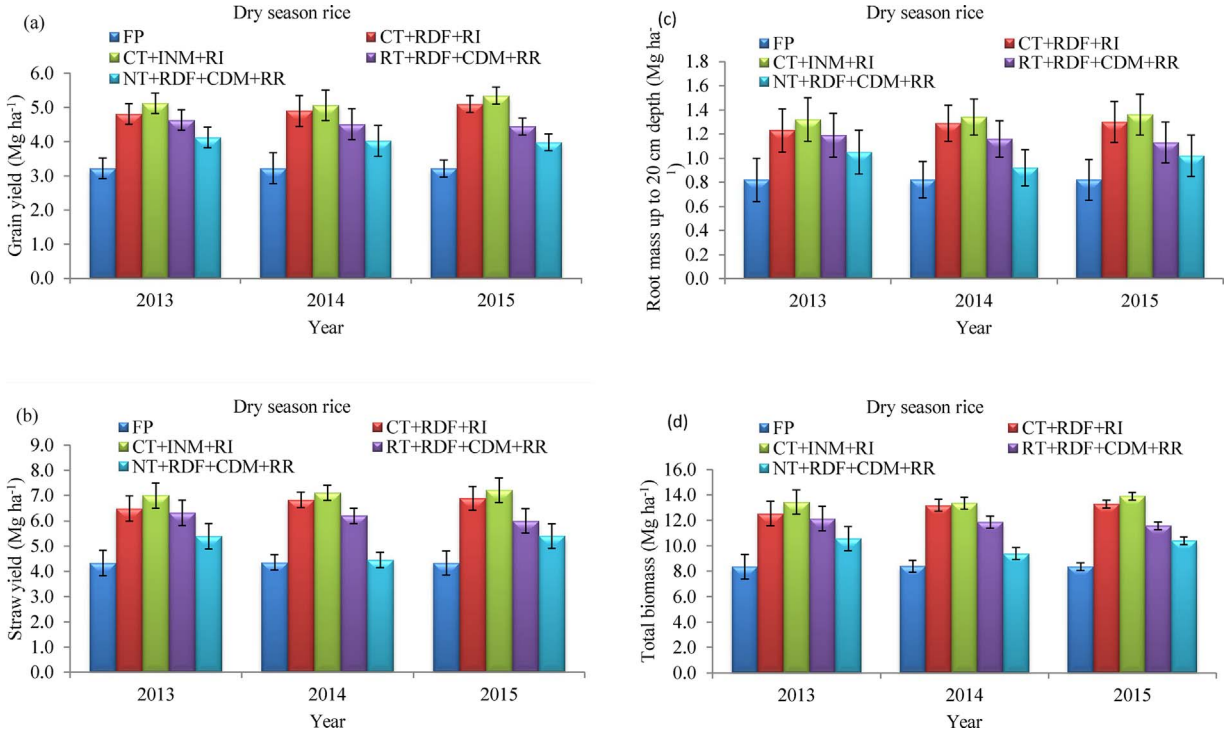


Fig. 3. Effect of tillage and nutrient management on grain (a), straw (b), root (c) and biomass (d) yield of dry season rice under rice–rice system (the vertical error bars indicate LSD at $p = 0.05$%).

FP: Conventional tillage (CT) + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation (RI); CT + RDF + RI: CT + 100 kg N, 18 kg P and 33.3 kg potassium (K) ha^{-1} (RDF) + 30% RI; CT + INM + RI: CT + integrated nutrient management (INM) comprising 25% N through FYM and 75% N & rest of P & K(after deducting contributions from FYM) through inorganic fertilizer + 30% RI; RT + RDF + CDM + RR: Reduced tillage (RT) + RDF + cellulose decomposing micro-organism (CDM) + 30% residue retention (RR); NT + RDF + CDM + RR: No-till (NT) + RDF + CDM + 30% RR; FP: farmer's practice; RDF: recommended dose of fertilizer.

throughout the crop growing season because of the integrated use of organic, inorganic and biofertilizers (Yadav et al., 2013b). The latter also increases SOC and improve soil structure and enhance the edaphic conditions (Das et al., 2014). Whereas application of CDMs under RT and NT with RDF did not increase the yield over that of CT + RDF + RI, treatments involving RT + RDF + CDM + RR and NT + RDF + CDM + RR produced 40.6 and 25.5% higher average grain yield over that of FP. In this context, only a few studies on conservation tillage (RT and NT) under DR have been conducted in India (Pandey and Velasco, 1999; Naresh et al., 2014, 2016). Most of the previous research related to CA involved the rice–wheat system, with special emphasis on wheat (Hobbs, 2007; Jat et al., 2009, 2013). While some experiments on effects of CA on RRS have been conducted in China and Bangladesh, results of these studies are contradictory. Several researchers in Bangladesh (Haque, 2009; Islam et al., 2014; Haque et al., 2016) reported no significant yield differences between RT and CT treatments. Several experiments in China (Peng et al., 2009; Huang et al., 2015; Mi et al., 2016) reported decrease in rice yield during the dry season, while others (Sun et al., 2010; Xue et al., 2015; Huang et al., 2015, 2016; Ma et al., 2016) reported increase in rice yield under conservation tillage irrespective of residue removal or retention. A meta-analysis of data from experiments in China indicated that grain yield responses to CA varied among regions, because of differences in climatic and edaphic factors (Huang et al., 2015). Further, CA effects on grain yield also vary with soil pH (Huang et al., 2015), and yield under NT can decrease by 2.8% in soils with pH < 6.0 (Gregory, 2012). The soil of present study had also lower pH (< 5.5). Huang et al. (2016) reported low rice yield under NT due to poor nutrient uptake and reduced biomass production in rice. These may also be the reason of low rice yield under RT + RDF + CDM + RR and NT + RDF + CDM + RR than the CT + INM + RI.

3.3. Effects on system yield and biomass productivity

System yield of grain, straw, root mass, and total biomass (grain + straw + root mass up to 20 cm soil depth) were also affected by tillage, residue and nutrient management practices (Table 4). The rice grain yields of the RRS ranged from 6.4 to 10.5 Mg ha^{-1} across treatments and years. However, impacts of treatments on system productivity were different than those recorded under individual WR and DR. The system grain, straw, root mass and total biomass yield over three consecutive years was 80–90% more under improved alternative management practices than that under FP. However, response of system varied among the improved management practices. The highest system productivity of RRS was achieved under T_3, which was statistically at par with that of T_4 and T_2. Similar trends were observed in the straw, root and total biomass production of system as that for the grain yield. Increased productivity of RRS under conservation tillage (RT/NT) with integrated nutrient and residue management has also been reported in other regions of world (Haque, 2009; Islam et al., 2014; Huang et al., 2015; Haque et al., 2016), but effects differed among regions because of differences in climatic and edaphic factors (Huang et al., 2015). The data of the present study demonstrate that cultivation of RRS under T_3 was the most suitable for the NER. However, further studies are needed to identify strategy to improve the productivity of DR under NT/RT under RRS for adoption CA in the NER.

3.4. Effects on biomass, carbon and nitrogen recycling

The amount of biomass added in soil, C and N recycling also differed among treatments, because biomass production (grain + straw + root mass upto 20 cm soil depth) was affected by tillage, nutrient and residue management (Table 5). With retention of residues by 30%, the root mass was assessed to 20 cm soil depth for each crop during the entire experiment for recycling of biomass, C and N. In addition, C and

Table 4
Effect of tillage and nutrient management on grain, straw, root and biomass yield of rice–rice system.

Treatment	Grain yield (Mg ha^{-1})			Straw yield (Mg ha^{-1})		
	2013	2014	2015	2013	2014	2015
T_1	6.44^{d}	6.55^{d}	6.42^{d}	8.88^{d}	9.05^{d}	8.86^{d}
T_2	9.23^{b}	9.51^{b}	9.57^{b}	12.72^{b}	13.10^{b}	13.19^{b}
T_3	9.93^{a}	9.90^{a}	10.46^{a}	13.80^{a}	14.11^{a}	14.51^{a}
T_4	9.73^{a}	9.63^{b}	9.65^{b}	13.51^{a}	13.42^{b}	13.35b
T_5	8.44^{c}	8.47^{c}	8.43^{c}	11.39^{c}	10.75^{c}	11.80^{c}

Treatment	Root mass yield (Mg ha^{-1})			Total biomass (Grain + straw + root) Mg ha^{-1}		
	2013	2014	2015	2013	2014	2015
T_1	1.67^{d}	1.69^{d}	1.66^{d}	16.99^{d}	17.29^{d}	16.94^{d}
T_2	2.39^{ab}	2.46^{a}	2.47^{ab}	24.34^{b}	25.07^{b}	25.23^{b}
T_3	2.58^{a}	2.64^{a}	2.71^{a}	26.31^{a}	26.65^{a}	27.68^{a}
T_4	2.53^{a}	2.50^{a}	2.49^{a}	25.77^{b}	25.55^{b}	25.49^{b}
T_5	2.17^{c}	2.09^{c}	2.20^{c}	22.00^{c}	21.31^{c}	22.43^{c}

T_1: Conventional Tillage (CT) + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation (RI) + farm yard manure (FYM) 5 Mg ha^{-1} once in two years in wet season rice (WR) and CT+ 40 kg N and 9 kg P ha^{-1} + 30% RI in dry season rice (DR); T_2: CT+ 80 kg N, 18 kg P and 33.3 kg potassium (K) ha^{-1} (RDF) + 30% RI in WR and CT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF) + 30% RI in DR; T_3: Reduce tillage (RT) + integrated nutrient management (INM) comprising 25% N through FYM and 75% N & rest of P & K(after deducting contributions from FYM) through inorganic fertilizer + 30% residue retention (RR) in WR and CT + INM + 30% RI in DR; T_4: RT + Improved plant nutrient management (IPNM) comprising 25% N through green leaf manure (GLM) + 60 kg N, 9 kg P, 17 kg K, 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} + cellulose decomposing micro-organism (CDM) + 30% RR in WR and RT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF) + CDM + 30% RR in DR; T_5: No-till (NT) + IPNM + CDM + 30% RR in WR and NT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF)+ CDM + 30% RR in DR; FP: farmer's practice; RDF: recommended dose of fertilizer; INM: integrated nutrient management; IPNM: improved nutrient management.
Within a Column, data for each parameter followed by same letter are not significantly different according to LSD ($p = 0.05\%$).

Table 5
Effect of tillage and nutrient management on biomass, carbon and nitrogen input, bulk density, soil organic carbon and nitrogen concentration, pools, accumulation, sequestration, carbon retention efficiency, microbial biomass carbon and dehydrogenase activity under rice–rice system.

Treatment	Biomass added in soil (through 30% RI/RR + RM Mg ha^{-1})	Total biomass added in soil (Mg ha^{-1})	Carbon added in soil (through 30% RI/RR + RM Mg ha^{-1})	Total carbon added in soil (Mg ha^{-1})	N added in soil (through 30% RI/RR + RM kg ha^{-1})	Total N added in soil (kg ha^{-1})
T_1	13.06^{d}	23.06^{d}	5.63^{d}	7.64^{d}	31.52^{d}	281.6^{e}
T_2	19.02^{b}	19.02^{c}	8.20^{b}	8.20^{b}	45.91^{b}	585.9^{a}
T_3	20.66^{a}	32.66^{a}	8.90^{a}	11.32^{a}	49.86^{a}	541.9^{b}
T_4	19.60^{a}	22.00^{b}	8.45^{a}	8.77^{b}	47.31^{a}	535.2^{c}
T_5	16.64^{c}	19.04^{c}	7.17^{c}	7.49^{c}	40.17^{c}	528.1^{d}

Treatment	BD (Mg m^{-3})	Soil organic carbon (g kg^{-1})	Total organic carbon pool (Mg ha^{-1})	Cumulative carbon sequestration (Mg ha^{-1})	Carbon sequestration rate (kg ha^{-1} year^{-1})	CRE (%)
T_1	1.33^{c}	10.42^{c}	20.77^{c}	0.42^{d}	141.6^{c}	5.59^{c}
T_2	1.33^{c}	10.42^{c}	20.78^{c}	0.50^{d}	144.7^{c}	5.19^{c}
T_3	1.31^{a}	11.01^{a}	21.63^{a}	1.30^{a}	427.9^{a}	10.98^{b}
T_4	1.32^{b}	10.89^{ab}	21.57^{a}	1.20^{b}	404.8^{a}	13.80^{a}
T_5	1.33^{c}	10.71^{b}	21.36^{a}	1.02^{c}	336.5^{b}	13.47^{a}

Treatment	Total soil nitrogen (g kg^{-1})	Total nitrogen pool (Mg ha^{-1})	Cumulative nitrogen sequestration (kg ha^{-1})	Nitrogen Sequestration rate (kg ha^{-1} year^{-1})	SMBC (μg g^{-1} dry soil)	DHA (μg g^{-1} dry soil)
T_1	1.019^{b}	2.03^{c}	18.0^{b}	6.0^{b}	245.0^{c}	5.88^{c}
T_2	1.025^{b}	2.04^{b}	29.7^{b}	9.9^{b}	234.5^{c}	6.18^{c}
T_3	1.050^{a}	2.06^{a}	47.6^{a}	15.9^{a}	314.1^{a}	10.48^{a}
T_4	1.041^{a}	2.05^{a}	46.2^{a}	15.4^{a}	307.4^{b}	9.80^{b}
T_5	1.033^{a}	2.06^{a}	45.1^{a}	15.0^{a}	304.4^{b}	9.49^{b}

T_1: Conventional Tillage (CT) + 40 kg nitrogen (N) and 9 kg phosphorus (P) ha^{-1} + 30% residue incorporation (RI) + farm yard manure (FYM) 5 Mg ha^{-1} once in two years in wet season rice (WR) and CT + 40 kg N and 9 kg P ha^{-1} + 30% RI in dry season rice (DR); T_2: CT + 80 kg N, 18 kg P and 33.3 kg potassium (K) ha^{-1} (RDF) + 30% RI in WR and CT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF) + 30% RI in DR; T_3: Reduce tillage (RT) + integrated nutrient management (INM) comprising 25% N through FYM and 75% N & rest of P & K(after deducting contributions from FYM) through inorganic fertilizer + 30% residue retention (RR) in WR and CT + INM + 30% RI in DR; T_4: RT + Improved plant nutrient management (IPNM) comprising 25% N through green leaf manure (GLM) + 60 kg N, 9 kg P, 17 kg K, 2 kg Boron (B) and 5 kg zinc (Zn) ha^{-1} + cellulose decomposing micro-organism (CDM) + 30% RR in WR and RT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF) + CDM + 30% RR in DR; T_5: No-till (NT) + IPNM + CDM + 30% RR in WR and NT + 100 kg N, 18 kg P and 33.3 kg K ha^{-1} (RDF)+ CDM + 30% RR in DR; FP: farmer's practice; RDF: recommended dose of fertilizer; INM: integrated nutrient management; IPNM: improved nutrient management; RM: root mass: SMBC: soil microbial biomass carbon; DHA: dehydrogenase activity.
Within a Column, data for each parameter followed by same letter are not significantly different according to LSD ($p = 0.05$%).

N were also added through input of FYM, GLM and fertilizer as per the respective treatment. Therefore, amount of biomass, C and N recycled through 30% residue + root biomass (20 cm soil depth) varied among the treatment. The highest biomass, C and N were recycled in T_3 plots through 30% residue + root biomass (20 cm soil depth) due to higher straw and root mass production. Total biomass and C recycled followed the trend of the addition of biomass and C through 30% residue incorporation + root biomass. However, the amount of N recycled was the highest in T_2 plots. Other researchers have also reported higher C and N recycling potential from retention of rice residue and with application of FYM and GLM (Dobermann and Witt, 2000) and rice residue and weed biomass (Das et al., 2008, 2014) under RT in lowland rice compared to that grown under a CT system.

3.5. Effects on soil bulk density

Tillage, nutrient and crop residue management significantly affected soil ρ_b at 0–20 cm soil depths (Table 5). By the end of the study, soil under T_3 recorded lower ρ_b than that under other treatments. However, soil ρ_b did not differ significantly among other treatments compared with that of the initial value. Continuous use of FP (T_1) and CT had no effect on soil ρ_b probably because of the incorporation of 30% of residues. Positive effects of the retention of crop residues on soil ρ_b in the surface 10 cm depth have been widely reported (Bhattacharyya et al., 2008; Govaerts et al., 2009; Mi et al., 2016). Puddling increases soil ρ_b immediately below the plow layer due to destruction of soil aggregates, reduction in porosity by filling of macropores with finer soil particles, and the direct physical compaction caused by the tillage implements (Sharma et al., 2003; Gathala et al., 2011). The data presented herein suggest that RT in both WR and DR crops may have relatively positive effect on soil ρ_b. In contrast, other studies have reported higher ρ_b under NT at 0–5 cm depth compared to that under CT in cropping systems other than RRS (Gathala et al., 2011; Huang et al., 2012; Jat et al., 2013). Management- induced changes in soil ρ_b depend on texture, SOM content, tillage type and intensity, and cropping system (Sharma et al., 2003; Kharub et al., 2004). Any change in soil ρ_b in the present study may have been masked by its clay loam texture and the fact that soil samples were obtained from the middle of 0–20 cm upper layer. Soil sampling at smaller depth increments should be preferred to detect ρ_b variability in the surface layer.

3.6. Effects on total soil organic carbon

The SOC concentration of 0–20 cm depth after completing three consecutive cycles of RRS was significantly affected by tillage, nutrient and residue management (Table 5). The SOC concentration increased from 2.1 to 7.9% under T_1 and alternative improved management practices. The data indicate that T_1 maintained a consistently higher SOC concentration in soil, which increased by 2.1% over the initial SOC concentration. Other researchers have documented that traditional rice farming practices can produce comparable SOC with an average increase of 2.2% over the initial value (Ramesh and Chandrasekaran, 2004). In the present study, the maintenance of SOC under T_1 might be due to incorporation of 30% of rice residues of each crop and application of 5 Mg FYM once in every two year. Several researchers (Mandalet al., 2004; Ramesh and Chandrasekaran, 2004; Lou et al., 2011; Chen et al., 2017) have also reported that the RR/RI and application of organic manure increased the SOC concentration in soils under rice. Further, paddy soils play a key role in total C sequestration and a sink for global C (Pan et al., 2004; Mi et al., 2016). In the present study, T_4 and T_5 increased SOC by 2.8–5.7% over T_1. However, cultivation of RRS under T_3 recorded the highest SOC concentration than that in soils under T_2 and T_1 treatments. The increase in SOC concentration might be due to the production of higher residues and root mass as well as addition of FYM (Zhou et al., 2016). The results demonstrated that CA (RT and NT) with RR/RI + improved nutrient management (INM/IPNM) enhances the SOC content in soil under RRS. After four years of tillage and straw retention practices in China, significant differences in SOC concentration (0–40 cm) were observed in the order of NT > RT > CT (Shang-Qi et al., 2013). In India, several researchers (Mandal et al., 2004; Ramesh and Chandrasekaran, 2004, Mandal et al., 2008; Das et al., 2014) have also reported that CA with organic manure and RR/RI increased the SOC in paddy soils. Further, the integrated use of organics with inorganic fertilizers enhances the productivity and appreciably increases the SOC over that in the control treatment (Nath et al., 2016).

Similar to SOC concentration, SOC pool in paddy soils was also increased under T_3 over that of the antecedent, T_1, and T_2 treatments. However, there were no significant differences in SOC under T_3 with those under T_4 and T_5. A total amount of 1.30 Mg C ha^{-1} was accumulated under soils of T_3 over three years. The rate of SOC sequestration ranged from 141.6 kg ha^{-1} yr^{-1} under T_1 to 427.9 kg ha^{-1} yr^{-1} under soils of T_3. Adoption of RT/NT system with INM/IPNM and retention/incorporation of 30% rice residues sequestered three times more SOC in soil system under RRS. The variation in SOC pool and sequestration were attributed largely to C addition through recycling of organic sources (FYM/GLM), tillage intensity (RT/NT), crop residues retention (30% residue), production of root mass, nutrient-use pattern, soil texture and prevailing ecosystem (Singh et al., 2015; Mi et al., 2016). Enhancement in SOC pool of the paddy soils can improve soil quality and mitigate global warming. Moreover, CT practice can enhance SOC and N mineralization by incorporating crop residues, disrupting soil aggregates, and increasing aeration, hence causing a reduction in SOC pool (Das et al., 2014; Xue et al., 2015; Huang et al., 2016). As a consequence, NT farming has been widely recommended as an alternative to CT for enhancing SOC sequestration in soils (Bessam and Mrabet 2003; Blanco-Canqui and Lal 2008; Das et al., 2014). Some studies have reported that NT practices can increase the SOC pool compared with that in soil under CT in paddy-based agro-ecosystems (Sun et al., 2010; Xue et al., 2015), and the application of rice residues with inorganic fertilizer may also enhance SOC sequestration in the double-rice cropping system in India (Bhattacharyya et al., 2012).

CRE is an important measure of the amount of C applied and retained in soil (Bhattacharyya et al., 2012). In the present study, CRE ranged from 5.2 to 13.8% across the treatments. The lowest CRE was observed in T_2 (5.2%) and the highest in T_4 (13.8%) under RRS while the highest amount of C (11.3 Mg ha^{-1}) was added under T_3. This trend in CRE indicates that the adoption of RT/NT systems under RRS is a better option for C sequestration than that of the CT-based systems (Bhattacharyya et al., 2012), and might be due to the slow decomposition of the applied biomass (Mi et al., 2016).

3.7. Effects on total soil nitrogen

TSN concentration, total nitrogen stocks (TNS), total nitrogen accumulation (TNA) and total soil nitrogen sequestration (TSNS) were also significantly affected by the tillage, nutrient and residue management after completion of three RRS cycles (Table 5). Trends almost similar to that of SOC were observed for TSN, TNS, TNA and TSNS after three years. However, cultivation of RRS under T_3 recorded significantly the highest value of TSN (1.05 N g kg^{-1}), TNS (2.06 Mg ha^{-1}), TNA (47.6 kg ha^{-1}) and TSNS (15.9 kg ha^{-1} $year^{-1}$) compared with those under T_2 and T_1, but did not show significant differences with those under T_4 and T_5. The data presented herein show that TSN, TNS, TNA, and TSNS increased significantly with the decrease in the tillage frequency and intensity along with the application of organic manure (FYM/GLM) and retention/incorporation of crop residues. In China, Xue et al. (2015) reported that the TN concentrations significantly decreased with the increase in the tillage frequency and intensity with the residue-retained treatments, and followed the order of NT > RT > CT at 0–5 cm depth. The CT practice is known to enhance SOC and N mineralization by incorporating crop residues,

disrupting soil aggregates, and increasing aeration, hence causing a reduction in TSN and N pools (Huang et al., 2016). As a consequence, NT farming has been widely recommended as an alternative to CT for enhancing TSN and N sequestration in soils (Xue et al., 2015; Huang et al., 2016).

3.8. Effects on soil microbial biomass carbon and dehydrogenase activities

SMBC and DHA ranged from 234.1 to 314.1 μg g^{-1} dry soil and 5.88–10.48 μg TPF g^{-1} dry soils, respectively, across the treatments. The SMBC concentration was significantly ($p = 0.05$) higher under T_3 (314.1 μg g^{-1} dry soil) and was the minimum under T_2 (234.5 μg g^{-1} dry soil) than those recorded in all other treatments. However, the SMBC concentrations under T_4 and T_5 were statistically similar (Table 5). Similarly, the DHA activity was also the maximum under T_3 (10.48 μg TPF g^{-1} dry soil) and the minimum under T_1 (5.88 μg TPF g^{-1} dry soil).

The SMBC and enzymes are soil quality indicators because of their relevance to soil biology, rapid response to changes in soil management, and ease of measurement (Dick et al., 1996). In the present study, increasing tillage intensity reduced SMBC and DHA. Soils under T_4 and T_5 had significantly higher SBMC and DHA compared to that under T_2 and T_1. The SMBC and DHA under T_3 were more than that under T_2 by 33.9 and 69.6%, respectively. Pandey et al. (2014) concluded that higher DHA in soils under NT and RT was due to larger proportions of SMBC than that in soil under CT. Higher activity of soil DHA and SMBC under RT and NT compared to that under CT has also been reported by other researchers (Mina et al., 2008; Ghosh et al., 2010; Pandey et al., 2014). The accumulation of crop residues on the soil surface is due to the enrichment of SOM in the surface layer and relatively higher microbial activity in soil under NT and RT (Mathew et al., 2012; Chen et al., 2017).

The concentration of SMBC is indicative of the soil's ability to store and cycle nutrients and SOM (Dick, 1992), and it also plays an important role in physical stabilization of aggregates (Franzluebbers et al., 1999). Consequently, SMBC is an important indicator of soil quality, and is also closely related to soil fertility (Das et al., 2014). The rate of biomass-C input from plant residues is a predominant factor controlling the amount of SMBC (Campbell et al., 1997). The continuous and a uniform supply of biomass-C is an energy source for microorganisms (Das et al., 2014). Thus, RT/NT with INM and crop residues are necessary for improving soil quality particularly in the depleted soils of NEH region (Boulal et al., 2008).

3.9. Effects on agroecosystems services

Agro-ecosystems provide food, forage, bioenergy, environmental equity and other essential functions for human wellbeing and nature conservancy (Lal, 2014). These systems also support and influence a variety of eco- system services, e.g., food and nutritional security, carbon sequestration, soil and water conservation and quality, biodiversity, mitigating greenhouse gases emission and other cultural services (Sagie and Ramon, 2015). Depending on management practices, agriculture production systems have both positive and negative impacts on these services. However, adoption of BMPs like conservation tillage, residue management, and INM may improve the agro- ecosystem services. In the present study, the adoption of T_3 increased the system productivity (economic yield) of RRS, improved soil quality (by lowering the ρ_b), and increased C and N pools, DHA and SMBC. The soil environment harbors the greatest biodiversity, has a high C pool, and may provide essential ecosystem services that impact soil nutrient cycling, water transfer, and crop growth, contributing to sustainable soil productivity (Roger-Estrade et al., 2010). In the present study, the T_3 had shown the potential of climate change mitigation as it had sequestered ~1.6 Mg CO_2 ha^{-1} yr^{-1}, which was higher than those under the T_1 (0.5 Mg CO_2 ha^{-1} yr^{-1}). Thus, T3 will contribute to climate change mitigation.

4. Conclusions

The data presented support the hypothesis that conservation tillage with the judicious management of nutrient and crop residue enhances the system productivity, and increases C and N sequestration in soils of RRS. The results discussed also support the following conclusions:

- Conservation tillage (RT/NT) with improved plant nutrient management and 30% residue retention enhanced grain yield by 51.1–52.2% over existing farmer's practices, and also sequestered SOC by 1.30 Mg C ha^{-1} in soils of RRS over three years.
- The rate of SOC sequestration increased from 141.6 kg ha^{-1} $year^{-1}$ under farmer's practice to 427.9 kg ha^{-1} $year^{-1}$ under soils of CA. In addition, SOC/TSN concentrations, pool, CRE, SMBC, DHA were also increased under CA with INM/IPNM and residue retention.
- The adoption of RRS under CA with INM/IPNM and effective residue recycling had the potential to sequester ~1.6 Mg CO_2 ha^{-1} yr^{-1} in paddy soils, which may help in mitigation of climate change.
- Thus, cultivation of RRS under CA with INM/IPNM and effective residue recycling is recommended for enhancing the system productivity and sequestration of C and N in paddy soil
- Further studies are needed to improve the productivity of DR under NT/RT under RRS for adoption CA in the NER Region.

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